Assignment 1 – Group 15

Roleplay Requirements Lists:

Patron Roleplay

regarding ability for books
* The system shall show the availability of a given book
* The system shall show my current account status, the status of every book book I have (ie if its late, and by how much), and the due date for each lent item without asking the front desk
* The system shall allow me to reserve an available book and hold a currently lent book get ahold of it book at its earliest availability
* The system shall allow book renewals if there are no holds for the book
regarding late returns
* The system shall offer a grace period for returning books for 3 days before any punishment is sent
* The system shall provide daily reminders via email to notify of upcoming due dates for lent items
* The system shall allow users to notify the library of possible late returns, in order to notify that the late return is possible
* The system shall allow a representative/automated message from the library to contact me if I am late to return a book on the last day of the grace period
regarding login
* The system shall default the library card number as the login for the library interface
additional requirements I think are important
* The system shall keep track of and show the history of my checkout and returns, sorted by most recent transactions

Librarian Roleplay

#Searching/Lookup
*The system shall let staff search for books by title, author, or by scanning a barcode and shall return results for similar matches

*The system shall let staff search for members using their name, library card or phone number
*The system shall show the status of each book when searched for (available, checked out, over due)
*The system shall show overdue books when a members account is opened
*The system shall look up books and return results within 3 seconds
*The system shall be able to search through a collection of 4000 books and 1000 members
#Fines
*The system shall fine anyone with an overdue book starting the day after the due date
*The system shall restrict anymore checkouts from member with overdue books until any outstanding fines are paid in full
*The system shall notify the member through email or a phone call that they have an overdue book that needs to be returned
*The system shall be capable of changing the daily fine amount
*The system shall be capable of having fines paid off in full or partially
#Checkout/Returns
*The system shall let staff complete a checkout in 3 steps: identify the member, scan or enter each book, then confirm
*The system shall let staff check out several books to one member in a single transaction.
*The system shall let staff renew a loan for one more loan period, unless another member has a hold on that book
*The system shall update a copy's status within 5 seconds of it being checked out or returned
*The system shall track each physical copy of a title separately, using a unique barcode for each copy.
#Ease of use
*The system shall display all main features staff need (Checkout, Return, Renew, a Book/Member search bar, and the Overdue Items list) on a single home screen
*The system shall allow the barcode to be scanned without the need to click onto the search bar

Librarian Director Roleplay (Joint)

regarding logging of users and books:
* The system shall keep track of and show the history of all book checkouts
* The system shall record the most popular books
* The system shall record all users and all active users
* The system shall be low maintenance

* The system shall be easy and cheap to deploy and keep running
regarding late fees
* The system shall keep track of late fees and offer grace periods for the returns of all members
* The system shall allow for growth future on for additional items like ebooks and inter-library loans in the future
* The system shall allow for a grace period for a few days, may offer a fine for repeated late returns, but may be replaced with other measures

Group Consolidation

#	Requirements	Source	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes / Resolution
1	The system shall show the availability of a given book	Patron	Pass	Pass	Pass	Pass	Pass	Pass	
2	The system shall show my current account status, the status of every book book I have (ie if its late, and by how much), and the due date for each lent item without asking the front desk	Patron	Pass	Pass	Pass	Pass	Pass	Pass	
3	The system shall allow me to reserve an available book, hold a currently lent book, and receive a book on hold at its earliest availability	Patron	Pass	Pass	Pass	Pass	Pass	Pass	
4	The system shall offer a grace period for returning books for 3 days, and offer no financial punishment for late returns	Patron	Fail	Pass	Pass	Pass	Pass	Pass	Conflicts with the librarian’s requirement of having a fee when the book is due

5	The system shall provide daily reminders via email to notify of upcoming due dates for lent items	Patron	Pass	Pass	Fail	Pass	Pass	Pass	Would need to implement an automatic email system to notify
6	The system shall allow users to notify the library of possible late returns, in order to notify that the late return is possible	Patron	Pass	Pass	Fail	Pass	Pass	Pass	Would need to integrate an email system for the library and its librarians in order to respond to the requests
7	The system shall allow a representative/automated message from the library to contact me if I am late to return a book on the last day of the grace period	Patron, Librarian	Pass	Fail	Fail	Pass	Pass	Pass	It is ambiguous of what the message should say, and the system of matching cards to emails/other contact information is required
8	The system shall default the library card number as the login for the library interface	Patron	Pass	Pass	Pass	Pass	Pass	Fail	Using the library card number as the login is not safe for security reasons, and may be hard to evolve afterwards
9	The system shall let staff search for members using their name, library card or phone number	Librarian	Pass	Pass	Fail	Pass	Fail	Pass	Need to implement a system that matches library ids to other contact information
10	The system shall let staff search for books by title, author, or by scanning a barcode and shall return results for partial matches	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	

11	The system shall show overdue books when a members account is opened	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
12	The system shall look up books and return results within 3 seconds	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
13	The system shall be able to search through a collection of 3500 books and 1000 members	Librarian	Pass	Pass	Pass	Pass	Pass	Fail	Needs to account for a larger increase in members/books
14	The system shall show the status of each book when searched for (available, checked out, over due)	Librarian	Pass	Pass	Fail	Pass	Pass	Pass	Does not mention lost/stolen or books that are on hold
15	The system shall fine anyone with an overdue book starting the day after the due date	Librarian	Fail	Pass	Pass	Pass	Pass	Fail	Conflict with patron and director, does not offer a grace period, No amount or cap.
16	The system shall restrict anymore checkouts from member with overdue books until any outstanding fines are paid in full	Librarian	Fail	Fail	Pass	Pass	Pass	Pass	Does not comply with the grace period put on books past due date, are these books on grace period overdue or not
17	The system shall be capable of changing the daily fine amount	Librarian	Fail	Pass	Pass	Pass	Pass	Pass	Conflict with patron not wanting fines
18	The system shall be capable of having fines paid off in full or partially	Librarian	Fail	Fail	Fail	Pass	Pass	Pass	Assuming fines exist(still in dispute), unclear whether system takes payments or only records

									them, also unsure about payment method
19	The system shall let staff complete a checkout in 3 steps: identify the member, scan or enter each book, then confirm	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
20	The system shall let staff check out several books to one member in a single transaction.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
21	The system shall let staff renew a loan for one more loan period, unless another member has a hold on that book	Librarian, Patron	Pass	Fail	Fail	Fail	Pass	Pass	Does not mention how long a loan period is or if they can renew once in total or can renew multiple times
22	*The system shall update a copy's status within 5 seconds of it being checked out or returned	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
23	The system shall track each physical copy of a title separately, using a unique barcode for each copy.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
24	The system shall display all main features staff need (Checkout, Return, Renew, a Book/Member search bar, and the Overdue Items list) on a single home screen	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	

25	The system shall allow the barcode to be scanned for checkout, returns and searches without the need to click onto the search bar	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
26	The system shall keep track of and show the history of all book checkouts, in reverse chronological order	Librarian Director, Patron	Pass	Pass	Pass	Pass	Pass	Pass	
27	The system shall record the most popular books	Librarian Director	Pass	Fail	Pass	Pass	Pass	Pass	Most popular books may mean most checkout, but likely wants to also sort by most popular books by genre
28	The system shall record all users and all active users	Librarian Director	Pass	Pass	Pass	Pass	Pass	Pass	
29	The system shall be low maintenance	Librarian Director	Pass	Fail	Pass	Pass	Pass	Pass	“Low maintenance” may mean a lot of things like server size or redundancy of the data
30	The system shall be easy and cheap to deploy and keep running	Librarian Director	Pass	Fail	Fail	Pass	Pass	Pass	Ambiguous to say what is cheap to deploy and run, and the requirement itself requires securing other details like hardware in order to be confirmed

31	The system shall keep track of late fees and offer grace periods for the returns of all members	Librarian Director	Pass	Pass	Pass	Pass	Pass	Pass	
32	The system shall allow for growth future on for additional items like ebooks and inter-library loans in the future	Librarian Director	Pass	Pass	Pass	Pass	Pass	Pass	
33	The system shall allow a fine for repeated late returns, but may be replaced with other measures	Librarian Director	Fail	Fail	Pass	Pass	Pass	Pass	Ambiguous of the punishment used for late returns, if its fines or other punishments, and would require rewrites to support the features

AI Requirement Consolidation

The AI is ChatGPT, and the requirements are produced by the model when scenario brief and told to generate a requirements list
** note that the Source of these requirements is not specified by the model. It is instead labelled by the group

#	Requirements	Source**	Conflict	Ambiguity	Completeness	Consistency	Testability	Evolvability	Notes / Resolution
1	The system shall maintain records of the library's books.	Director	Pass	Pass	Pass	Pass	Pass	Pass	

2	The system shall maintain records of library members.	Director	Pass	Pass	Pass	Pass	Pass	Pass	
3	The system shall record which member has borrowed each book.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
4	The system shall allow staff to look up books.	Librarian	Pass	Fail	Pass	Pass	Pass	Pass	Look up by what term? Keyword, author, title?
5	The system shall allow staff to look up members.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
6	The system shall track whether a book is currently borrowed or available.	Patron	Pass	Pass	Fail	Pass	Pass	Pass	Only allows for two states, what if book is on hold, lost/damaged
7	The system shall assist staff in identifying books that have not been returned on time.	Librarian	Pass	Pass	Pass	Pass	Pass	Pass	
8	The system shall maintain sufficient lending information to help identify books that may be missing.	Librarian	Pass	Fail	Pass	Pass	Pass	Pass	Ambiguous of what the stored lending information is
9	The system shall support approximately 3,000 books.	Director	Pass	Pass	Pass	Pass	Pass	Fail	Does not account for an increase in books/e-books
10	The system shall support approximately 800 active members.	Director	Pass	Fail	Pass	Pass	Pass	Fail	Does not account for an increase in members, what is considered an active member, do accounts that have been

									inactive for a long period get deleted
11	The system should allow staff to retrieve information quickly.	Librarian	Pass	Fail	Pass	Pass	Pass	Pass	Quickly is ambiguous, compared to the spreadsheet and paper many options are fast
12	The system should be simple to operate because it is intended for a small library branch.	Librarian	Pass	Fail	Pass	Pass	Pass	Fail	The notion of simple is ambiguous, and adding in more options may make the system harder to use
13	The system should be inexpensive to develop, deploy, and maintain because the library has a limited budget.	Director	Pass	Fail	Fail	Pass	Pass	Pass	It is ambiguous on what inexpensive for a system at this scale is, and what resources are already available for the branch
14	The system should replace the existing paper-card-catalog/shared-spreadsheet workflow without introducing unnecessary complexity.	Director	Pass	Pass	Fail	Pass	Pass	Pass	Replacement would require updating and storing the old system into the new one. This may be a problem itself.
15	Staff can add new books.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
16	Staff can create member accounts.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
17	The system keeps backups	Director	Pass	Fail	Pass	Fail	Pass	Pass	How often are the

									backups made? Where are they stored? Fails in completeness for the formatting not matching.
18	The system supports fines.	Director	Pass	Fail	Fail	Fail	Pass	Pass	Note that there is no conflict here because of the lack of shareholders views here, but it fails because it doesn't note how the fines are paid or how they are obtained. Fails in completeness for the formatting not matching.
19	The system notifies members about overdue items.	Director	Pass	Fail	Pass	Fail	Pass	Pass	How are they notified? Fails in completeness for the formatting not matching.
20	Staff can update/deactivate members.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
21	Staff can remove books.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
22	Each physical book copy has a unique identifier.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
23	Staff can edit book information.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not

									matching.
24	Staff can check books back in.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
25	The system supports reservations/holds.	Director	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
26	Staff can mark books as lost, damaged, or missing.	Librarian	Pass	Pass	Pass	Fail	Pass	Pass	Fails in completeness for the formatting not matching.
27	The system generates reports.	Director	Pass	Fail	Pass	Pass	Pass	Pass	What is contained in the report? And who is reading these reports?

Resolutions for the Requirements:

Here we note what requirements we have selected, but requires a revision in order to resolve the failed attribute:

Group Consolidation :

Original Requirement	Revised Requirement	Solution
The system shall provide daily reminders via email to notify of upcoming due dates for lent items AND The system shall allow users to	The system shall provide daily reminders via email to notify of upcoming due dates for lent items AND The system shall allow users to notify the library of possible late returns, in	Clarifies that the information of the member can store email and phone numbers, allowing these requirements to function

notify the library of possible late returns, in order to notify that the late return is possible	order to notify that the late return is possible. The system shall allow for users to have make an account with their library card number, their name, and optionally an email and phone number for contacting them	
The system shall default the password for their account as the library card number as the login for the library interface	The system shall default the password for their account as the library card number as the login for the library interface, with options to change it available	Resolves evolability concerns with giving login a password that is explicitly changeable
The system shall be able to search through a collection of 3500 books and 1000 members	The system shall be able to search through a collection of 3500 books and 1000 members, with ability to also handle more	Increase evolvability as it specifies the bounds as a minimum to achieve
The system shall show the status of each book when searched for (available, checked out, over due)	The system shall show the status of each book when searched for (available, checked out, over due, lost, stolen, or on hold)	Completes the options for states the book can take by adding in lost, stolen, or on hold
The system shall restrict anymore checkouts from member with overdue books until any outstanding fines are paid in full	The system shall be able to restrict anymore checkouts from member with overdue books until any outstanding fines or any associated penalty for the late returns are resolved	Clears ambiguity
The system shall let staff renew a loan for one more loan period, unless another member has a hold on that book	The system shall let staff renew a loan for 7 days starting at the time the renewal is made and can be renewed continuously unless another member has a hold on that book	Resolves problems with ambiguity, consistency, and testability by specifying the length of the renewal and the ability to repeatedly renew the same book

The system shall record the most popular books	The system shall record and sort the the most popular books in the library and the most popular books in each genre	Removes ambiguity of what is tracked and increases usefulness to track per-genre
The system shall be low maintenance	The system shall be low maintenance and not require active supervision of the system	Resolves the ambiguity of the system being low maintenance as the ability to be able to run without supervision
The system shall be easy and cheap to deploy and keep running	The system shall be easy and simple to run, being able to run inside of the library's computers itself	Removes the ambiguity of the system being low maintenance by specifying the low-resource nature of the system, as it can fit on a library computer
The system shall offer a grace period for returning books for 3 days, and offer no financial punishment for late returns The system shall keep track of late fees and offer grace periods for the returns of all members The system shall fine anyone with an overdue book starting the day after the due date		This part of fees/penalties of overdue books remains contentious across the shareholders, with the patron and director being for no-fees while the librarians suggest implementing them. This is left as a conflict for the parties to be resolved, and for now the implementations should allow for capabilities with/without fees being tracked and paid for.

AI Consolidation:

Original Requirement	Revised Requirement(s)	Reasoning
The system shall maintain sufficient lending information to help identify books that may be missing	The system shall maintain sufficient lending information of the member who last obtained the book and the date to help identify books that may be missing.	fixes ambiguity of what is stored
The system should replace the existing paper-card-catalog/shared-spreadsheet workflow without introducing unnecessary complexity	The system should replace the existing paper-card-catalog/shared-spreadsheet workflow without introducing unnecessary complexity. The system shall allow for migration from paper-card-catalog/shared-spreadsheet tools with integration of scanning and porting these files over into the system	resolves problems with how migration can be handled between the older and newer systems
Staff can add new books	The system shall allow staff to add new books	formatting issues for consistency
Staff can create member accounts	The system shall allow staff to create member accounts	formatting issues for consistency

Staff can update/deactivate members	The system shall allow staff can update/deactivate members on their behalf	formatting issues for consistency
Staff can remove books.	The system shall allow staff to remove books	formatting issues for consistency
Staff can edit book information.	The system shall allow staff to edit book information	formatting issues for consistency
Staff can check books back in.	The system shall allow staff to check books back in	formatting issues for consistency
The system keeps backups	The system shall store monthly backups of the transactions performed in the last month in a local database, and includes the current status of every account with a loan/hold	removes ambiguity of what is stored, how often, and where it's stored. Also fixes formatting for consistency
The system generates reports.	The system generates reports of the total and types of transactions performed in the system, like loan times, overdue durations, books added, books returned, etc.	specifies what is contained in the report

Comparison between the Group and AI Requirements

- What did the AI's list miss that your role-play surfaced (especially any stakeholder conflict)?
The AI's list was solely based on the brief overview and missed a majority of what the stakeholder role play uncovered. From the patron side, they requested for more self service features, including showing their current account status, the status of every book they have (i.e. if it's late, and by how much) and the due date for each lent item without asking the front desk. The librarian was also a lot more specific than the AI. They wanted to search for books by title, author or barcode and for members by name, library card or phone number. The director wanted a history of all checkouts, the most popular books and a count of active users, as well as room to grow for ebooks and inter-library loans. The AI only says "the system generates reports" and doesn't mention growth. One of the key points that the AI missed is the conflict with fines and overdue books. The librarian wanted a small daily fine starting the day after the due date to decrease the amount of missing/overdue books, and wanted to block more checkouts until fines are paid in full. While the patrons were strongly against any charges due to late books, and would rather a grace period to be given. The director is on the fence about fines, and is ok with some repercussions against chronically late returns but would like to keep the public image of the library positive and not punitive to lower-income patrons, suggesting a grace period of a few days with a possible fine only for repeated late returns. The AI's list only states that the system supports fines, which picks the librarian's side without noting any disagreement, when this should have been flagged as an open conflict for the director to decide.

- What did the AI invent or assume that wasn't grounded in any actual stakeholder need?
The AI assumed that the director would want backups but the director only wanted the system to be low maintenance and cheap to run. It also assumed that the system shall support fines, which is supported by the librarian stakeholders but is in conflict with the patrons and director. It also assumed some standard admin functions, such as adding,editing, and removing books.

- Where did the two requirements lists agree?
The two lists generally agree on the capabilities of the system, being able to search members, books, track history of the books, and being cheap and easy to operate. They produce similar looking requirements in these regards, but due to the lack of shareholders studies the AI is not given, its list is more generalized than specific as the AI is afraid of making such conclusions, ie. looking up a book in 3 seconds.

- Did the AI identify any useful requirements that your group had not initially considered?
With our focus on facilitating the shareholder’s requirements as given by the rolecards, we had missed out on key abilities of the system like creating/updating/removing members and staff accounts, checking in back books, migration and data backup plans. How the system fundamentally operates were missed out during our roleplay, and so the set of requirements we produced may have satisfied the needs of the shareholders, in practice the system would fail to operate.

Final refined requirements list

#	Requirements
1	The system shall show my current account status, the status of every book I have (ie if its late, and by how much), and the due date for each lent item without asking the front desk
2	The system shall allow me to reserve an available book, hold a currently lent book, and receive a book on hold at its earliest availability
3	The system shall allow for users to have make an account with their library card number, their name, and optionally an email and phone number for contacting them
4	The system shall provide daily reminders via email to notify of upcoming due dates for lent items AND The system shall allow users to notify the library of possible late returns, in order to notify that the late return is possible
5	The system shall allow users to notify the library of possible late returns, in order to notify that the late return is possible
6	The system shall default the password for their account as the library card number as the login for the library interface, with options to change it available
7	The system shall let staff search for books by title, author, or by scanning a barcode and shall return results for partial matches
8	The system shall show overdue books when a members account is opened
9	The system shall look up books and return results within 3 seconds
10	The system shall be able to search through a collection of 3500 books and 1000 members, with ability to also handle more
11	The system shall show the status of each book when searched for (available, checked out, over due, lost, stolen, or on hold)
12	The system shall be able to restrict anymore checkouts from member with overdue books until any outstanding fines or any associated penalty for the late returns are resolved

13	The system shall be capable of changing the daily fine amount
14	The system shall be capable of handling fines paid at the front desk with the librarian
15	The system shall be capable of having fines paid off in full or partially
16	The system shall let staff complete a checkout in 3 steps: identify the member, scan or enter each book, then confirm
17	The system shall let staff check out several books to one member in a single transaction.
18	The system shall let staff renew a loan for 7 days starting at the time the renewal is made and can be renewed continuously unless another member has a hold on that book
19	The system shall update a copy's status within 5 seconds of it being checked out or returned
20	The system shall track each physical copy of a title separately, using a unique barcode for each copy.
21	The system shall display all main features staff need (Checkout, Return, Renew, a Book/Member search bar, and the Overdue Items list) on a single home screen
22	The system shall allow the barcode to be scanned for checkout, returns and searches without the need to click onto the search bar
23	The system shall keep track of and show the history of all book checkouts, in reverse chronological order
24	The system shall record and sort the the most popular books in the library and the most popular books in each genre
25	The system shall record all users and all active users
26	The system shall be low maintenance and not require active supervision of the system
27	The system shall be easy and simple to run, being able to run inside of the library's computers itself
28	The system shall keep track of late fees and offer grace periods for the returns of all members
29	The system shall allow for growth future on for additional items like ebooks and inter-library loans in the future

30	The system shall support a configurable grace period for overdue items and configurable consequences for late returns.
31	The system shall record which member has borrowed each book.
32	The system shall allow staff to look up members.
33	The system shall assist staff in identifying books that have not been returned on time.
34	The system shall maintain sufficient lending information of the member who last obtained the book and the date to help identify books that may be missing
35	The system should replace the existing paper-card-catalog/shared-spreadsheet workflow without introducing unnecessary complexity
36	The system shall allow for migration from paper-card-catalog/shared-spreadsheet tools with integration of scanning and porting these files over into the system
37	The system shall allow staff to add new books
38	The system shall allow staff to create member accounts
39	The system shall allow staff can update/deactivate members on their behalf
40	The system shall allow staff to remove books
41	The system shall allow staff to edit book information
42	The system shall allow staff to check books back in
43	The system shall store monthly backups of the transactions performed in the last month in a local database, and includes the current status of every account with a loan/hold
44	The system generates reports of the total and types of transactions performed in the system, like loan times, overdue durations, books added, books returned, etc.

Proof of the Pre-Work Checklist:

Link to the public repository: https://github.com/DaLawnChair/cisc327_roleplay
Screenshot of the main page and collaborators list:



